Simple gear trains

National 5 · Mechanisms & structures · Answers

Answers and marking guidance

Working is shown for each answer. Other valid methods should be credited.

1 State which gears you used as the driver and the driven, and calculate the velocity ratio.

- Driver 40 T, driven 8 T
- VR = 8 ÷ 40 = 0.2 (1 : 5): the output turns 5 times faster

2 Now change your gear train in Yenka to give the biggest decrease in speed. State the gears and calculate the velocity ratio.

- Driver 8 T, driven 40 T
- VR = 40 ÷ 8 = 5 (5 : 1)

3a) Which gear rotates in the same direction as A?

- Gear C

3b) Calculate the velocity ratio from A to B, A to C and A to D.

- A to B: 10 ÷ 50 = 0.2
- A to C: 25 ÷ 50 = 0.5
- A to D: 100 ÷ 50 = 2

3c) A rotates at 500 revs min^{-1}. Calculate the speeds of B, C and D.

- B: 500 ÷ 0.2 = 2500 revs min^{-1}
- C: 500 ÷ 0.5 = 1000 revs min^{-1}
- D: 500 ÷ 2 = 250 revs min^{-1}

4a) Calculate the velocity ratio of the gear system.

- B is an idler, so VR = C ÷ A = 30 ÷ 20 = 1.5

4b) Calculate the output speed.

- 250 ÷ 1.5 = 167 revs min^{-1}

4c) State the direction of rotation of gear C.

- Anticlockwise (the same as A)

5a) How many teeth does gear B have?

- 40 teeth (B turns half as many times, so it has twice the teeth)

5b) What is the velocity ratio?

- VR = 40 ÷ 20 = 2 (2 : 1)

5c) Shaft A rotates at 60 revs min^{-1}. At what speed does shaft B rotate?

- 60 ÷ 2 = 30 revs min^{-1}

5d) Shaft A rotates anticlockwise. In which direction does shaft B rotate?

- Clockwise

6a) Calculate the velocity ratio.

- VR = 45 ÷ 15 = 3 (the idler doesn't change it)

6b) Calculate the speed of the largest gear.

- 1800 ÷ 3 = 600 revs min^{-1}

6c) The driver turns anticlockwise. In which direction does the output turn?

- Anticlockwise (an idler makes them turn the same way)